

Wild Realms - An Interactive Wildlife Education Simulation

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For our next implementation, Wild Realms will expand upon the original prototype by introducing interactive modes. On the biome selection page, users will have the option to switch between two modes:

1. Exploration Mode (default)
2. Challenge Mode

Scenario “Challenge Mode”

We’ve already implemented most of the Exploration Mode. This will be the default mode that loads when the user clicks “Continue” and is sent to the biome selection page. In this mode, the user can select a biome and explore the species that live there.

Currently, when a user selects a biome in the selection page, it takes them to a satellite view of the biome (which we implemented with the google maps API). The user can already view species associated with each biome and see key information about them, such as their diet and conservation status.

To finish the Exploration Mode, we will add two buttons to each biome, an “Immersive” button and a “Satellite” button. These will be the two ways you can view each biome. The “Satellite” view option will be what we already have which is a satellite map of the biome which we got from google maps API. The second option will be the “Immersive” view option. When chosen, this mode will display the biome as a 3D-rendered environment complete with 3D animals carrying out their natural behaviors. Within each biome in “Immersive mode”, there will be a virtual tour guide (likely displayed as pop-up text) that explains the significance of each observed behavior and the species’ role in the ecosystem.

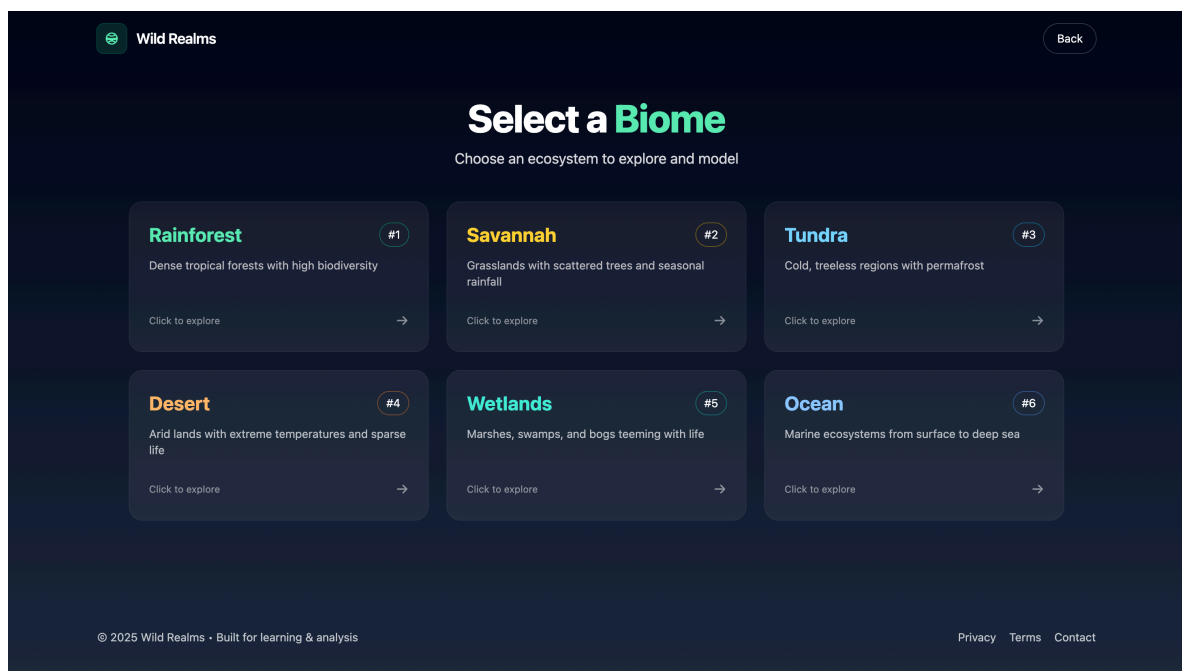
Additionally, there will be a “Start Game” button in each biome. In this mini-game, users can select and learn about different species, and once they discover a certain number of them, they will earn an achievement for completing that biome.

The second mode we will implement is Challenge Mode. In this mode, when a user selects a biome, they will encounter a specific environmental threat that affects that ecosystem. For example, if the user selects the ocean biome, they will experience a pollution simulation, where they must make a series of choices to mitigate the damage and restore balance to the environment. The gameplay focuses on decision-making. If the user makes all the right decisions, they will successfully save that biome and earn an achievement for completing it in

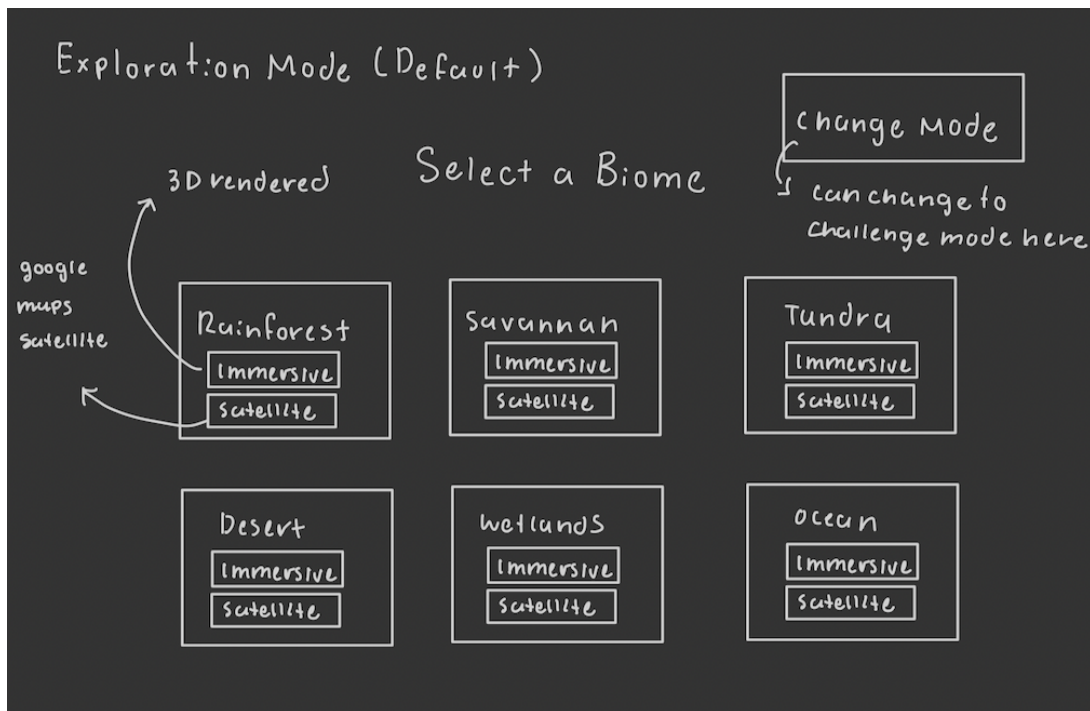
Challenge Mode. As a side note, the satellite view will not exist for the Challenge mode because it doesn't make sense for the implementation since Challenge mode will be a game in the 3D rendered version of each biome.

In addition to these new interactive modes, the scenarios page for each species will also be updated to include calculations based on real data. For example, for the African Zebra, if a user selects "Human Activity," the simulation will use real-world research to estimate a realistic decrease in the population and show that change in the population bar for that animal.

Current Selection Page:



With our second implementation:



There will be a change mode button on the Challenge mode here as well to switch back to exploration mode.

